ICT COURSE COVERAGE

COURSE COVERAGE: Class 8

Sr. no	Theme	Competency	Objectives	Title	Tools	Periods		
						Demons--tration	Hands on & Assessment	Total Periods
1	Data Representation & Processing-04 (Spreadsheet)	Students will be able to develop the skills for creating effective and accurate visualizations of data sets.	▪ To create spread--sheets and represent data using graphs, interpret graphs and identify data patterns ▪ To sort and filter data	▪ Representing data using graphs. • Line chart, • bar chart, • pie chart ▪ Sorting data ▪ Filtering data	Open Office Soft-ware Suite	3	6	9
2	Learn-to-Code-03	Students will be able to develop creative thinking, critical think-ing, problem solving and integration of application with real life scenarios	▪ To create email ids and pass-word ▪ To create and represent infor-mation using an app	▪ Creating Authorised email login credentials and manag-ing pass-words ▪ Selecting an app layout. ▪ Creating user Interface ▪ Adding elements like buttons, la-bels, images etc. ▪ Overview of Block coding of app ▪ Testing app on phone ▪ Creating apk	MIT App inventor https:// builder. andromo. com/	4	8	12
3	Digital Well-being	Students will be able to de-velop healthy online work-ing method-ology and learn how to maintain bal-ance between the real and virtual world.	▪ To learn about Cyber ethics ▪ To learn about balancing on-line and offline activities	▪ How to behave with people in vir-tual world. ▪ Health implication of excess screen-time ▪ Concept of a habit tracker. ▪ Methods for digital detox-ification	Class ac-tivities like Skit pre-sentation, RolePlays, poster mak-ing, Slogan writing etc.	2	4	6

4 .	Being Future Ready- 02	Students will be able to develop creative thinking, critical thinking and problem solving skills. Student will also be able to enhance curiosity and imagination·	▪ To learn basics and uses of Artificial Intelligence(AI) in real life. ▪ To know the various Domains of Artificial Learning ▪ To understand AI Ethics and bias ▪ To understand the impact of AI on Sustainable Development Goals to develop responsible citizenship	▪ Introduction to Artificial Intelligence ▪ Applications of AI in real life ▪ Artificial Intelligence. Domains ▪ Artificial Intelligence Ethics and bias ▪ Introduction to sustainable development goals .	▪ AI for oceans an free online Platform achieving SDG 14: Life Below Water https://code.org/oceans AI Domain activities: Data : Rock, Paper and Scissors https://rockpaper-scissors-ai.vercel.app **CV:** https://quickdraw.withgoogle.com/ **NLP :** Google assistant	5	14	19
5	Exhibition & Evaluation of Portfolio			Showcasing/ e-portfolio		0	4	4

Suggested Tools/Resources

Theme	Suggested Resources
Data Representation & Processing -04	Other tools : spreadsheets open and libreoffice **Link: https://www.office.org/en/openoffice/** Mobile Apps : WPS Office, Excel, sheets, office365 **CWSN:** We can use window inbuilt Accessibility tools like text to speech, alt text etc. **Google sheets :** Tools menu (spell check, personal dictionary, auto complete), Accessibility (screen reader support-for Braille support, Screen Magnifier) - FREE

Learn- to- Code- 03	Mobile Apps : MIT AI2 COMPANION **Link: https://play.google.com/store/apps/details?id=edu.mit.appinventor.aicompanion3** **CWSN :** Code Jumper (A Coding Language for the Blind and Visually Impaired | Science Times) **Link:** https://codejumper.com/
Digital Wellbeing	**Resources :** Training on Online Safety Portocal and Cyber Security. **Link: https://diksha.gov.in/explore-course/course/do_31353451906048000113** Skill Education Book **Link: https://cbseacademic.nic.in/skill-education-books.html**
Being Future Ready -02	Other tools Students to join AI Community initiated by CBSE **Link: https://aistudent.community** Students to join AI for ALL Community initiated by CBSE and Intel **Link: https://ai-for-all.in/#/home?ref=CBSE** **CWSN :** Supersense - AI for Blind, Supersense is the smartest assistive app that helps blind and visually impaired users to read, find objects, and explore places independently. It provides a set of digital eyes to make the physical world more accessible for the blind and low vision community. **Link: https://play.google.com/store/apps/details?id=com.mediate.supersense** Bear 71 by Jam3 and National Film Board of Canada - Bear 71 is the Planet Earth of the future. It's a nature documentary blended with a VR experience that looks great in a headset or a browser. **Link: https://bear71vr.nfb.ca/** **Links :** **https://aidemos.microsoft.com/luis** **https://experiments.withgoogle.com/collection/ai** **https://code.org/** **https://teachablemachine.withgoogle.com/train** Free Self directed online courses which are available on the links provided **Note:** Teachers to give demo of AI activities like Face or object recognition, sound recognition and embed that code on free web portals, free self directed online courses which are available on the links provided.

Suggested Projects:

1. **Creating a mobile app:**
   Create a mobile app using tools of your choice with the theme Sustainable Development Goals. The students can choose from the following SDGs:
   Zero Hunger : SDG - Goal 2
   Zero Poverty : SDG - Goal 1
   These apps can be tested on any smartphone or on computer and links should be submitted.

2. **Use of AI:**
   Students will create Website using tools learnt in class 7 on various applications of Artificial Intelligence they have learnt in class 8.

REFERENCES

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- A Model Curriculum for ICT in Education by NCERT
- Position Paper on Curriculum, Syllabus & Textbooks by NCERT
- Secondary Curriculum 2015-16 Main Subjects Volume 1 by CBSE
- ICT curriculum for ICSE Schools
- ICT Curriculum for Rajasthan State Board Schools
- ICT Curriculum for Madhya Pradesh State Board Schools
- ICT Curriculum for Gujarat State Board Schools
- https://cbseacademic.nic.in/web_material/Curriculum22/publication/middle/AI_Curriculum_Facilitator_Handbook_VI_VIII.pdf
- Computer Science Curriculum for Schools, Model Curriculum and Teaching (Material for K-12 Indian Schools) Sri Sri Ravishankar Vidya Mandir (SSRVM)

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